Customer Churn Prediction and Segmentation-Based Retention Strategy Using Machine Learning and Explainable AI

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Name of the Company with address : NOT APPLICABLE – ACADEMIC PROJECT USING PUBLIC DATASETS

Abstract

Customer churn costs telecom and banking firms revenue, but a churn score alone does not say why a customer is at risk or what to do about it. This project builds a leakage-safe machine-learning pipeline that predicts which customers will leave, explains the reasons, groups customers by those reasons and recommends a retention action for each group. It is tested on two public datasets: Telco (7,043 customers, 26.5% churn) and a retail bank (10,000 customers, 20.4% churn).

The pipeline compares seven classifiers (logistic regression, random forest, SVM, XGBoost, LightGBM, CatBoost and an explainable boosting machine) by 5-fold cross-validation, with PR-AUC as the main metric because churners are a minority. XGBoost, a gradient-boosted decision-tree model, is tuned and kept as the lead model, and its decision threshold is chosen by expected cost from an explicit cost-benefit matrix instead of accuracy. SHAP values from out-of-fold predictions explain each prediction and are clustered with K-Means into customer segments, and DiCE counterfactuals suggest actionable changes. A Cox survival model on Telco estimates each customer's lifetime value, and a Streamlit demonstration ties the steps together.

On an untouched test set, XGBoost reaches PR-AUC 0.654 (Telco) and 0.718 (bank), and the cost-chosen threshold lowers expected cost by 8.7% and 8.8% compared with contacting nobody. SHAP-based segments separate churn better than clusters of raw features (churn-rate spread 0.654 against 0.498 in Telco and 0.808 against 0.191 in the bank). The survival model reaches a concordance index of 0.867, and a lifetime-value break-even rule saves 9.4%. Contract

commitment is the most stable lever in Telco. The cost figures are assumptions, because no campaign data exists, and all results describe associations, not causes.

Additional notes. XGBoost was kept as the lead model because no other model beat it by more than 0.01 PR-AUC on both datasets, and the explainable boosting machine, a glass-box model, matches it. The lifetime-value break-even rule saves 9.4% against contacting nobody, compared with 8.7% for the best single threshold. An error analysis shows that the churners the model misses look like loyal customers. Further checks qualify these results: the Cox proportional-hazards assumption holds only approximately, a neural survival model (DeepSurv, concordance index 0.868) is no better than Cox (0.867), and LIME broadly agrees with SHAP. False-alarm rates are higher for higher-churn groups (bank women, customers in Germany, Telco senior citizens), so these customers would receive more unnecessary offers.

1. Introduction

Customer churn directly reduces revenue, and keeping an existing customer costs far less than winning a new one. A company that can tell which customers are about to leave, and why, can spend its retention budget where it pays back.

Most churn studies stop at a prediction score. This project goes further in four steps: it predicts who will churn, explains the prediction with SHAP, groups customers by the reasons behind their risk, and recommends an action for each group, while counting the money an offer costs and the money a missed leaver costs.

1.1 Research questions

- **RQ1.** Which model predicts churn best across both a telecom and a banking dataset?
- **RQ2.** Does choosing the decision threshold by expected financial cost reduce cost compared with the default threshold of 0.5 and with no intervention?
- **RQ3.** Do segments built by clustering on SHAP values differ from segments built on raw features, and are they more useful for choosing retention actions?
- **RQ4.** Which small, feasible changes to a customer's actionable attributes would change a churn prediction?
- **RQ5.** Which churners does the model miss, and what do they have in common?

- **RQ6.** Does a customer-specific lifetime value, estimated with survival analysis, improve the cost-based decision compared with one fixed value for every customer?

1.2 Scope and guided enhancements

The topic was approved by the project guide, Dr. V. Swaminathan, together with four enhancements to a standard capstone: clustering on individual SHAP values ("SHAP-to-Segment"), a financial cost-benefit matrix in evaluation, DiCE counterfactual explanations, and a mandatory second dataset from the banking domain. After reviewing the results, the guide recommended four further additions: CatBoost and EBM in the model comparison, survival analysis on the Telco data, an error analysis of false negatives, and a working demonstration. The guide ranked model comparison and error analysis as the main items, survival analysis as an extension, and the Streamlit application as a demonstration.

1.3 Contributions

1. A leakage-safe, reusable pipeline that runs unchanged on two datasets from different industries.
2. A cost-based decision rule, with a sensitivity analysis of the cost assumptions.
3. SHAP-to-Segment clustering built on out-of-fold SHAP values, compared directly with raw-feature clustering and linked to counterfactuals and retention actions.
4. An error analysis of the churners the model misses.
5. A survival-analysis extension that gives each Telco customer a lifetime value.
6. A Streamlit demonstration of the full chain for a single customer.
7. Robustness checks: a proportional-hazards test, a neural survival model (DeepSurv), a LIME comparison with SHAP and a group-level check of false-alarm rates.

1.4 Structure of the report

Chapter 2 reviews related work. Chapter 3 describes the data. Chapter 4 explains the methods. Chapter 5 gives the main results, Chapter 6 the survival extension and Chapter 7 the demonstration. Chapters 8 and 9 discuss limitations and conclude.

2. Literature review

Eleven sources were reviewed. Each reference's title, authors, venue, year, pages and DOI were checked against publisher, arXiv or index pages before inclusion. The search was limited, not exhaustive.

2.1 Churn prediction in telecommunications and banking

Churn prediction is a well-studied problem in both telecommunications and retail banking. Verbeke et al. (2012) benchmarked a wide range of classifiers on eleven real telecom datasets and reported that a small number of variables suffices to predict churn accurately and that many classifiers perform comparably. In banking, Guliyev and Tatoglu (2021) found that XGBoost (Chen & Guestrin, 2016) outperformed other methods and used SHAP values to interpret it, while Brito et al. (2024) proposed a retail-banking framework combining feature engineering, class-balancing and XGBoost, validated on a dataset of more than three million customers. These findings are consistent with the present project, in which boosting models performed best, or tied for best, on both datasets.

2.2 Class imbalance and evaluation metrics

Churners are usually a minority, which makes plain accuracy misleading. Chawla et al. (2002) introduced SMOTE, which creates synthetic minority examples, and showed that combining over-sampling of the minority class with under-sampling of the majority class can improve classifier performance. However, Verbeke et al. (2012) found that oversampling generally did not improve churn-prediction performance significantly, and Elkan (2001) argued that changing class balance has little effect on some standard learners and that cost-sensitive decisions are better made by combining predicted probabilities with a cost matrix. Saito and Rehmsmeier (2015) showed that precision-recall plots are more informative than ROC plots when classes are imbalanced, because ROC plots can give a deceptive picture. This project therefore uses PR-AUC as its main metric and compares imbalance methods inside cross-validation folds.

2.3 Cost-sensitive and profit-driven evaluation

Because different errors have different business costs, statistical metrics alone can lead to suboptimal model selection. Verbeke et al. (2012) developed a profit-centric measure that finds

the most profitable fraction of customers to include in a retention campaign, and showed that it selects models and campaign sizes differently from statistical measures. Elkan (2001) formalised how a cost matrix and class probabilities yield the decision with the lowest expected cost. The present project follows this logic by choosing the decision threshold that minimises expected cost under explicit cost assumptions and by testing sensitivity to those assumptions.

2.4 Explainable machine learning

SHAP (Lundberg & Lee, 2017) assigns each feature an importance value for an individual prediction and unifies several earlier explanation methods under one framework with desirable theoretical properties. Lundberg et al. (2020) extended this to tree-based models with an exact, efficient algorithm, and showed that combining many local explanations can reveal global model structure and distinct population subgroups with shared risk characteristics. SHAP has been used for churn analysis in both telecom and banking, typically to identify global drivers such as contract type and tenure (El Attar & El-Hajj, 2026; Guliyev & Tatoglu, 2021).

2.5 Counterfactual explanations

Counterfactual explanations describe how a prediction could change if inputs were different. Mothilal et al. (2020) argued that useful counterfactuals must be feasible given a person's context and diverse, and proposed DiCE, which generates sets of diverse counterfactuals and allows constraints on which features may change. This motivates restricting changes to actionable features, as done here.

2.6 Segmentation and the research gap

El Attar and El-Hajj (2026) combined churn prediction, SHAP analysis and autoencoder-based segmentation on telecom data, and noted that studies combining ensemble models, explainable AI and unsupervised segmentation remain scarce. Lundberg et al. (2020) showed that explanations can reveal subgroups in medical data. In the sources reviewed for this project, no study was found that (i) clusters out-of-fold SHAP values of a churn model, (ii) compares the resulting segments with segments from raw features, (iii) links the segments to cost-optimal decision thresholds and to counterfactual levers, and (iv) tests the whole framework on both a telecom and a banking dataset. The contribution claimed here is this combination, not clustering

on SHAP values alone, and the statement reflects the limited search conducted, not an exhaustive review of the literature.

2.7 Summary of sources

Source	What it covers	Why it matters here
Lundberg & Lee (2017)	SHAP, a unified, game-theory-based way to attribute a prediction to features	Basis of the explanations
Lundberg et al. (2020)	Exact SHAP for tree models; combining local explanations to show global structure and subgroups	Basis of TreeExplainer; prior use of explanations to find subgroups
Mothilal et al. (2020)	DiCE: diverse, feasible counterfactuals	Basis of the counterfactual step
Chawla et al. (2002)	SMOTE oversampling for imbalanced data	One of the three imbalance methods
Saito & Rehmsmeier (2015)	Precision-recall plots are more informative than ROC under imbalance	Why PR-AUC is the main metric
Verbeke et al. (2012)	Profit-based evaluation of churn models; benchmark on 11 telecom datasets	Supports cost-based evaluation and the oversampling finding
Elkan (2001)	Cost-sensitive decisions use probabilities and a cost matrix	Basis of the threshold choice
Chen & Guestrin (2016)	XGBoost	The lead model
El Attar & El-Hajj (2026)	Telecom churn with SHAP and autoencoder-based segmentation	Closest related work

Source	What it covers	Why it matters here
Guliyev & Tatoglu (2021)	Banking churn, SHAP, XGBoost best	Banking evidence
Brito et al. (2024)	Retail banking churn framework, validated on a very large dataset	Banking context

3. Data

Both datasets are public Kaggle datasets, so no private customer data is used. Using two datasets from different industries tests whether the framework holds beyond one business context.

	Telco Customer Churn	Bank Customer Churn
Domain	Telecommunications	Retail banking
Customers	7,043	10,000
Columns (raw / after cleaning)	21 / 20	14 / 11
Churn rate	26.5%	20.4%
Training / test rows (80/20)	5,634 / 1,409	8,000 / 2,000

3.1 Cleaning decisions

- Raw files are never edited; cleaned versions are written by code to a separate folder.
- **Telco TotalCharges** was stored as text. Its 11 blank entries all belong to customers with zero tenure who had not yet been billed, and none churned. They were set to 0, the true amount billed so far. Dropping the 11 rows (0.16%) or filling with the mean were rejected, because the mean would invent a charge.
- **"No internet service" and "No phone service"** were merged into "No" in seven Telco columns. These labels occur only when the parent service is absent, so they duplicate another column and would split SHAP credit.
- Identifier columns were removed (customerID; RowNumber, CustomerId, Surname).
- The target is encoded 1 = churned, 0 = stayed, and named `churn` in both datasets.

- Bank EstimatedSalary is uncorrelated with every other column, so it is not used for cost assumptions.

3.2 Exploratory findings

Telco.

- Churn by contract: month-to-month 42.7%, one year 11.3%, two year 2.8%.
- Churn by tenure: 47% in months 0-12, 29% in 13-24, 20% in 25-48 and 9.5% in 49-72. The median tenure of leavers is 10 months against 38 for stayers, so customers are most at risk in their first year.
- Tenure and TotalCharges are strongly correlated (0.83), which is why TotalCharges cannot be used beside tenure in the survival models.

Bank.

- Germany 32% against about 16-17% in France and Spain; female 25% against male 17%. These are associations, not causes, and they raise a fairness question (Chapter 8).
- Number of products: one product 28%, two 8%, three 83% (n = 266), four 100% (n = 60). The last two groups are small.
- Inactive members 27% against 14% for active members. Holding a credit card has no effect (21% against 20%).
- Age is non-linear: 7.5% up to age 30, 12% at 31-40, 34% at 41-50, 56% at 51-60 and 25% above 60. The youngest customer is 18, so there are no minors.

These simple patterns are used later as a cross-check on the model explanations.

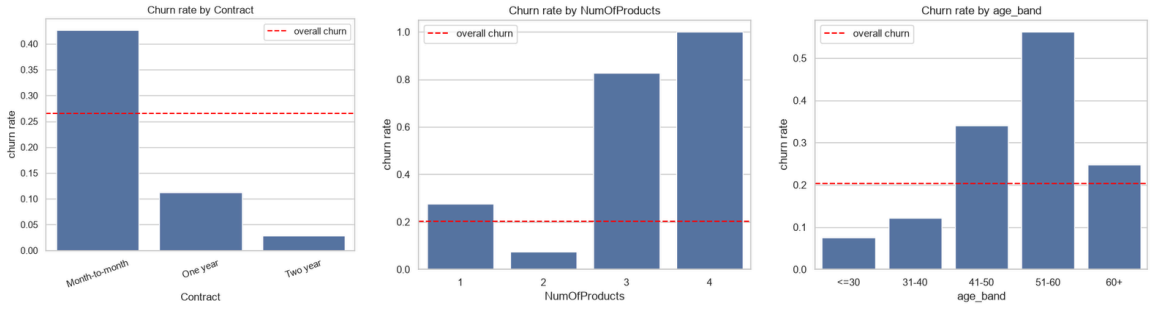


Figure 1. Churn rate by Telco contract type (left), by bank number of products (middle) and by bank age band (right).

4. Methodology

The same code runs on both datasets. Every choice below is recorded in a decisions log kept in the project repository.

4.1 Validation design

Each dataset is split 80/20 with stratification (random seed 42). The 20% test set is sealed and used once, for the final evaluation. All model selection uses 5-fold stratified cross-validation on the training data only. Preprocessing (one-hot encoding, scaling) and any resampling sit inside a pipeline, so they are fitted on the training part of each fold and cannot leak information from the validation part.

4.2 Models and metrics

Seven classifiers are compared: Logistic Regression (baseline), Random Forest, XGBoost, LightGBM, SVM with an RBF kernel (comparison only, no probabilities), CatBoost and the Explainable Boosting Machine (EBM), a glass-box additive model. The metrics are precision, recall, F1, ROC-AUC and PR-AUC. PR-AUC is the main metric because churners are the minority class and accuracy misleads: on the bank data, predicting that nobody churns already scores 79.6% accuracy.

4.3 Class imbalance

Class weights, SMOTE and random undersampling are compared, each applied only inside the training folds. Resampling mostly moves the decision threshold rather than improving the ranking, and it distorts predicted probabilities. Because the cost analysis needs honest probabilities, the final model is the unweighted XGBoost.

4.4 Cost-benefit matrix

A missed churner costs more than an unnecessary offer, so the threshold is chosen by expected cost. With V the value of a customer, offer the cost of one retention offer and s the probability that an offer retains a leaver:

Outcome	Cost
True positive (offer sent, customer was leaving)	offer + (1 - s) x V
False positive (offer sent, customer would have stayed)	offer
False negative (leaver missed)	V
True negative	0

Neither dataset has real campaign costs, so these are labelled assumptions: Telco V = 777 (twelve months of the mean monthly charge), offer = 78 (10% of V), s = 30%; bank (illustrative units) V = 1,000, offer = 100, s = 30%. The break-even probability is offer / (s x V), about 0.33. A sensitivity analysis varies s from 10% to 50% and the offer cost from 5% to 20% of V. Thresholds are chosen on out-of-fold training probabilities.

4.5 SHAP and SHAP-to-Segment clustering

1. SHAP values are computed out-of-fold, so every customer is explained by a model that never saw that customer.
2. SHAP values of the one-hot pieces of each original feature are summed into one value per original feature (valid because SHAP values are additive).
3. K-Means is run on this SHAP matrix; all values share one unit, so no rescaling is needed.
4. The number of clusters k is chosen with the elbow method, the silhouette score and stability across random starts (adjusted Rand index, ARI). The rule set in advance is the highest silhouette among k = 3 to 6 with stability above about 0.8, preferring the smaller k when scores are close.
5. For comparison, K-Means with the same k is run on the standardised raw features. The two segmentations are compared by their spread of churn rates and their agreement (ARI).
6. Segments are profiled with mean SHAP values (what drives risk) and typical feature values (who the customers are), then mapped to retention actions.

4.6 Counterfactual explanations (DiCE)

DiCE generates diverse counterfactuals for typical high-risk customers from the riskiest segments, with the random-search method. Only actionable features may change. For Telco these are contract, payment method, paperless billing, add-on services and the monthly charge (at most a 20% discount per customer). For the bank they are the number of products, active-member status and credit-card holding. Demographics, tenure, country, balance, credit score and salary stay fixed. A small wrapper makes the cost-optimal threshold act as DiCE's decision boundary.

4.7 Error analysis

The out-of-fold predictions are used to profile the false negatives: how many churners are missed, in which segments, with which tenure, contract or other characteristics, and how far their predicted probability lies below the threshold.

4.8 Survival analysis (Telco)

Tenure is the time variable and churn the event; customers who have not churned are right-censored. Tenure and TotalCharges are removed from the predictors. Four models are compared: Cox proportional hazards, Random Survival Forest (RSF), XGBoost-AFT (accelerated failure time) and DeepSurv (a small neural network trained with the Cox loss), scored by the concordance index (ranking quality) and the integrated Brier score (accuracy of the predicted survival probabilities over months 3 to 60). The Cox proportional-hazards assumption is checked with a rank-based Schoenfeld-residual test. A customer-specific value is defined as the monthly charge multiplied by the expected number of active months in the next 12, estimated from the Cox survival curve conditional on tenure so far.

4.9 Safeguards for validity

- All preprocessing and resampling happen inside cross-validation folds.
- The test set is used once, after all decisions are fixed.
- Thresholds and SHAP values used for clustering come from training data, out-of-fold.
- Random seeds are fixed (42), including the DiCE search, which is seeded through the global random generators.

- Tools: Python, pandas, scikit-learn, imbalanced-learn, XGBoost, LightGBM, CatBoost, interpret (EBM), SHAP, LIME, dice-ml, lifelines, scikit-survival, PyTorch, Streamlit, matplotlib.

4.10 Additional checks

Proportional hazards. The Cox model assumes that each feature multiplies a customer's risk by a constant factor over time. This was tested with a rank-based Schoenfeld-residual test (lifelines, ridge penalty 0.1), with a Bonferroni correction for the 21 tests.

DeepSurv. A small neural network (two hidden layers of 32 units, dropout 0.1, weight decay 0.001, 200 epochs, full-batch) was trained with the Cox partial likelihood in PyTorch and scored with the same 5-fold cross-validation as the other survival models. Like them, it was not tuned.

LIME. LIME fits a simple model around a single prediction. It was compared with SHAP on 100 randomly chosen flagged customers per dataset, using the share of the five most important features (on the one-hot encoded columns) that the two methods have in common.

Group check. On the out-of-fold predictions at the cost-optimal threshold, the flag rate, recall and false-alarm rate were computed for bank gender and geography and for Telco gender and senior-citizen status, to see whether errors fall unevenly across groups. As a simple mitigation test, the models were then retrained without the sensitive attributes (Telco gender and senior-citizen status; bank gender and geography) and compared on PR-AUC, expected cost and group error rates, each at its own cost-optimal threshold.

5. Results

5.1 Model comparison (RQ1)

No model beats XGBoost by more than 0.01 PR-AUC on both datasets, so XGBoost is kept as the lead model. Cross-validated PR-AUC on the training data:

Model	Telco PR-AUC	Bank PR-AUC
EBM	0.672	0.701

Model	Telco PR-AUC	Bank PR-AUC
CatBoost	0.667	0.711
XGBoost	0.666	0.701
Logistic Regression	0.661	0.469
Random Forest	0.659	0.684
LightGBM	0.656	0.702
SVM (RBF)	0.637	0.657

On Telco the top models lie within noise of each other, so complex models add little over logistic regression. On the bank data, boosting helps clearly (0.70 against 0.47 for logistic regression) because age and number of products act non-linearly. EBM matches XGBoost on both datasets while being directly interpretable, which suggests the churn signal is largely additive.

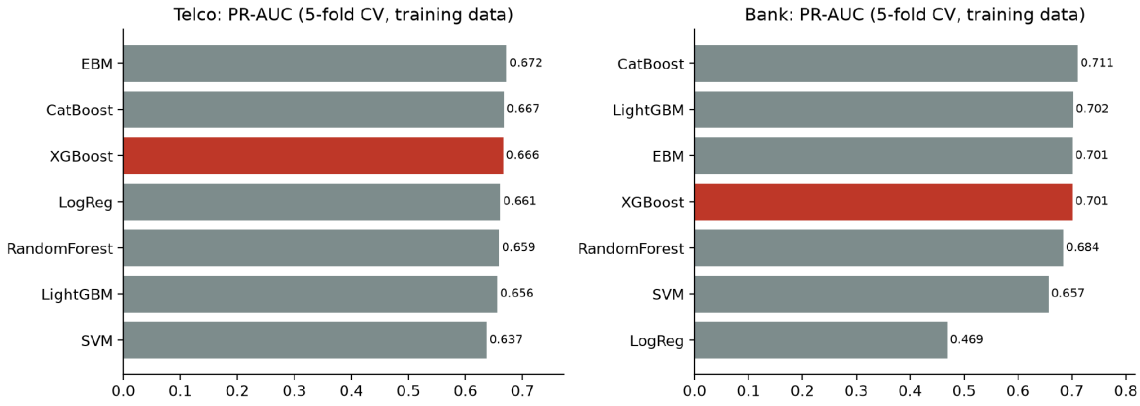


Figure 2. Cross-validated PR-AUC by model (5-fold, training data); XGBoost in red.

The rule fixed in advance was that another model replaces XGBoost only if it wins by more than 0.01 PR-AUC on both datasets. The largest gains are +0.006 (Telco, EBM) and +0.010 (bank, CatBoost, exactly at the margin), so XGBoost stays. This also keeps the SHAP, segment and counterfactual results valid, since they were built on XGBoost.

Class weights, SMOTE and undersampling raised recall (bank XGBoost 0.478 to 0.756 with undersampling) but lowered precision (0.753 to 0.483) and left ROC-AUC and PR-AUC roughly flat; they mostly shift the threshold. A 25-setting random search for XGBoost moved PR-AUC

from 0.666 to 0.671 on Telco and from 0.701 to 0.700 on the bank, within noise, so the original settings were kept.

5.2 Cost-based threshold and final test (RQ2)

On out-of-fold training probabilities the cost-optimal threshold is 0.36 for Telco and 0.32 for the bank, close to the break-even probability of 0.33. The cost curves are flat near their minimum (about 0.25 to 0.45), so the exact value is not critical. The test set was then used once:

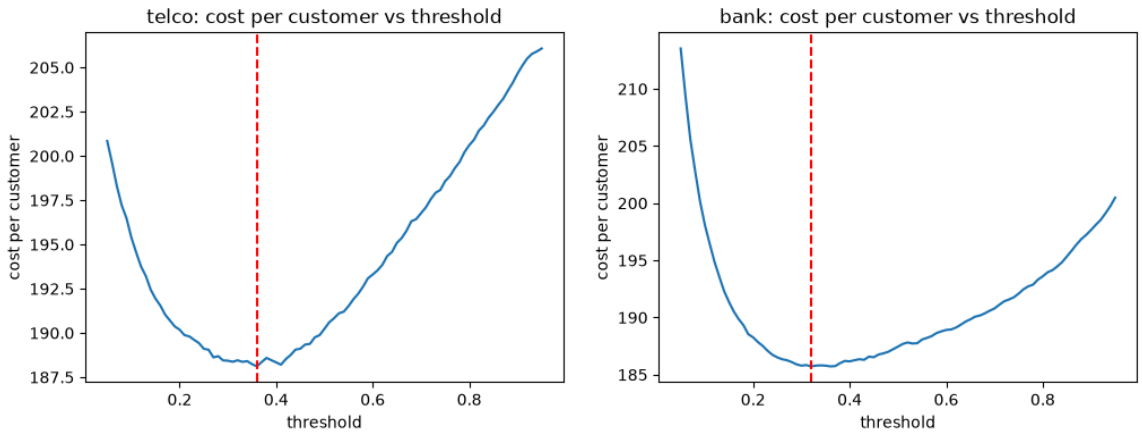


Figure 3. Expected cost per customer against the decision threshold (out-of-fold training probabilities).

	Telco	Bank
Threshold (from training CV)	0.36	0.32
Precision	0.569	0.616
Recall	0.703	0.641
F1	0.629	0.628
ROC-AUC	0.842	0.866
PR-AUC	0.654	0.718
Cost per customer at the chosen threshold	188.3	185.6

	Telco	Bank
Cost per customer at threshold 0.5	190.2	186.3
Cost per customer, contact nobody	206.2	203.5
Cost per customer, contact everybody	222.4	242.4

The cost-chosen threshold lowers expected cost by 8.7% (Telco) and 8.8% (bank) against contacting nobody, and by about 1% against the default 0.5. Test scores are close to the cross-validation scores, so there is no sign of overfitting. In every saving reported in this report, the baseline is contacting nobody, and the saving is (baseline cost − model cost) / baseline cost; the cost at the default 0.5 threshold is shown for comparison.

The sensitivity analysis shows that the value of the model depends on how well offers work. Across success rates of 10% to 50% and offer costs of 5% to 20% of V, the best threshold stays close to offer / (s x V) (from 0.05 to 0.95), and the saving against contacting nobody ranges from 0% (offers costly or ineffective) to about 35% (cheap, effective offers). The real success rate should therefore be measured with a pilot campaign. Because both datasets use the same cost ratios, their similar savings are partly by construction.

5.3 Explanations with SHAP

In Telco, contract type is the strongest driver of predicted churn, in line with the exploratory finding that month-to-month customers churn at 42.7% against 2.8% for two-year contracts. In the bank data, number of products and age are strong drivers, again consistent with the exploratory findings. SHAP values from the model fitted in cross-validation (out-of-fold) were stored for every training customer and used in the next section.

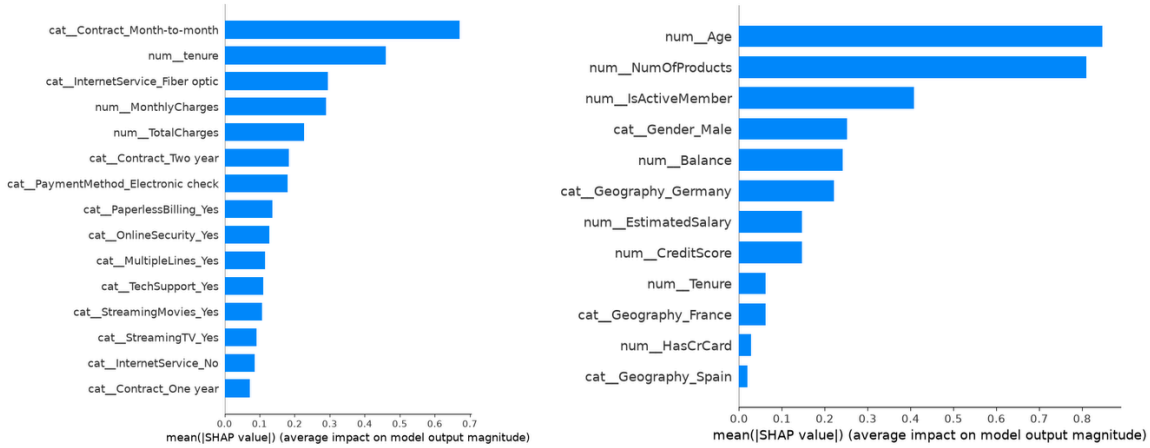


Figure 4. Global feature importance (mean absolute SHAP value), Telco (left) and bank (right).

As a check on the explanations, LIME (a local surrogate method) was compared with SHAP on 100 flagged customers per dataset. For each customer the five most important features from the two methods were compared. On average 3.3 of 5 features agreed in Telco (overlap 0.662) and 3.7 of 5 in the bank (0.732), and 89% and 96% of customers shared at least three. The two methods therefore broadly agree. Exact matches are not expected, because LIME fits a local linear approximation to the model while TreeSHAP computes exact attributions, and LIME was run on the one-hot encoded columns, so one category can appear as several features.

5.4 SHAP-based segments and retention strategy (RQ3)

The bank uses k = 4: its silhouette (about 0.36) ties with k = 5 and the elbow bends at 4. In Telco the silhouettes are low (0.23 to 0.26), which signals weak cluster structure; the pre-set rule would pick k = 3, but k = 6 was chosen because it separates risk and retention actions more finely (differences among k = 3 to 6 are small). A k = 3 run confirms this: its segments hold 2,335, 1,581 and 1,718 customers with churn of 5.7%, 44.0% and 38.9% (spread 0.383 against 0.654 for k = 6; ARI between the two groupings 0.457), because it merges the very high-risk new customers (67% churn) with longer-tenure customers at 38% to 44%. The segments are therefore a practical partition of a continuum, not sharply separated natural clusters.

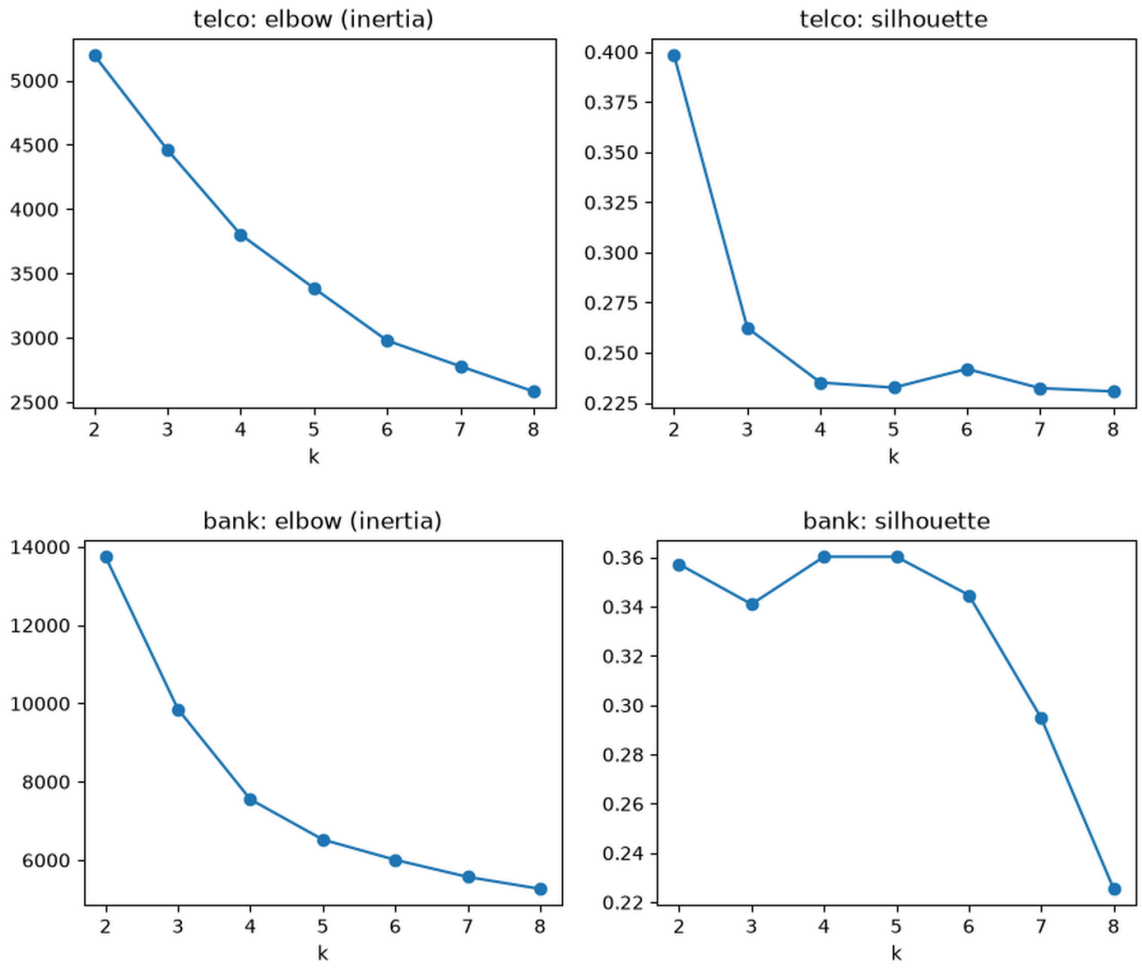


Figure 5. Elbow (inertia) and silhouette scores for k = 2 to 8, Telco (top) and bank (bottom).

Telco segments (k = 6, 5,634 training customers)

Segment	Customers	Churn	Who they are
4	934	67.1%	New (mean tenure 4 months), month-to-month, fibre, pay by electronic check
2	1,225	38.4%	Established (34 months), expensive (83 per month), month-to-month, fibre
0	943	24.3%	Short tenure (11 months), cheap (38 per month), month-to-month, mostly DSL

Segment	Customers	Churn	Who they are
5	942	13.5%	Long tenure (47 months), one-year contract, fibre
3	523	4.6%	Very long tenure (69 months), two-year contract, many add-on services
1	1,067	1.7%	Long tenure (43 months), two-year contract, cheap, few services

Bank segments (k = 4, 8,000 training customers)

Segment	Customers	Churn	Who they are
2	256	85.5%	Three or more products (mean 3.2), mean age 43
0	1,796	43.5%	Older (mean age 51), mostly one product (1.33)
3	2,869	16.9%	Younger (mean age 35), one product
1	3,079	4.7%	Younger (mean age 35), two products

SHAP-based segments separate churn better than clusters on raw features, and the two groupings largely disagree:

	Telco	Bank
Churn-rate spread, SHAP-based segments	0.654	0.808
Churn-rate spread, raw-feature segments	0.498	0.191
Agreement between the two (ARI)	0.23	0.10

This comparison needs one caution: SHAP values come from a model trained on churn labels, so SHAP-based segments are risk-aware by construction, while raw-feature clusters are not.

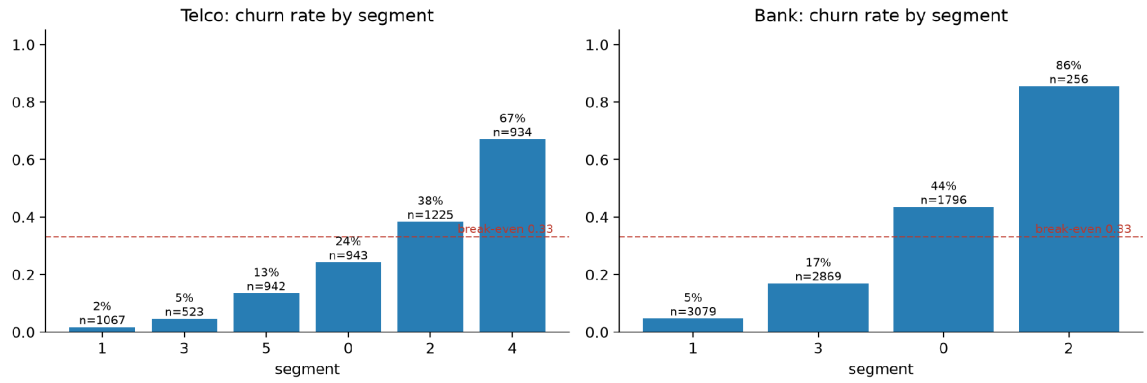


Figure 6. Churn rate by SHAP-based segment; the dashed line is the break-even probability (0.33).

Retention actions are aimed at segments whose average churn exceeds the break-even probability (about 0.33). The actions below are hypotheses to be tested in a pilot, not proven effects.

Dataset	Segment	Suggested action
Telco	4 (67%)	Offer a one-year contract with an incentive, encourage automatic payment, add an early onboarding call
Telco	2 (38%)	Offer a contract-commitment discount and a bundle of security and support add-ons
Telco	0 (24%)	Below break-even on average: offer a one-year contract only to customers whose own risk is above break-even
Telco	5, 3, 1	Low risk: no offer; protect service quality
Bank	2 (86%)	Review the experience of customers with three or more products through a relationship-manager call and a bundle review; do not simply remove products
Bank	0 (44%)	Run an activity and re-engagement programme and offer a second product
Bank	3, 1	Low risk: no offer; monitor

5.5 Counterfactual explanations (RQ4)

In brief: every Telco option moves the customer to a one- or two-year contract, so contract commitment is the stable lever, while the bank options concern the number of products and activity status. DiCE was run for one typical flagged customer from each of the two riskiest segments per dataset. A first Telco run let the monthly charge vary freely and produced unrealistic price cuts of 50% to 77%, so the final runs limit the discount to 20%.

- **Telco, segment 4** (customer 5037, risk 0.70, new month-to-month fibre customer): the saved options are a two-year contract with device protection added, a one-year contract with online security added, and a two-year contract with online backup added.
- **Telco, segment 2** (customer 5750, risk 0.51, established month-to-month fibre customer): a two-year contract with a discount of about 12%, a two-year contract with a discount of about 2%, and a one-year contract with online backup added.
- **Bank, segment 0** (customer 5341, one product, inactive, risk 0.63): becoming an active member, combined with a second product, a credit card, or both.
- **Bank, segment 2** (customer 2210, three products, risk 0.95): reducing to one or two products flips the prediction. This is read as a signal about the experience of multi-product customers, not as advice to remove products.

The main Telco lever is contract commitment, which is consistent with SHAP and with the exploratory findings: every Telco option moves the customer to a one- or two-year contract, and the other changes are small add-ons or discounts. The search is seeded (seed 42 for the global random generators), so the options repeat exactly between runs; a different seed gives different add-ons and discounts (earlier unseeded runs returned, for example, a two-year contract with a discount of about 19%), but in every run the contract was the stable lever. Options that switch a service off are model artefacts, not recommendations. Only four customers were analysed in detail, and becoming active is an outcome that needs its own actions.

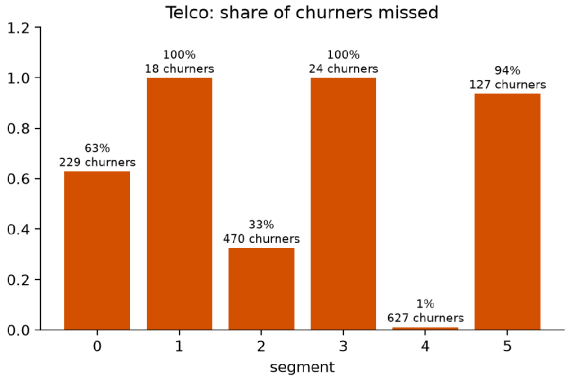
5.6 Error analysis of false negatives (RQ5)

Out-of-fold predictions at the cost-optimal threshold show which churners the model misses.

	Telco	Bank
Churners in training data	1,495	1,630
Caught	1,030	1,036
Missed	465	594
Recall	0.689	0.636

The misses are concentrated in the lower-risk segments:

Dataset	Segment	Churners	Share missed
Telco	4	627	1.1%
Telco	2	470	32.6%
Telco	0	229	62.9%
Telco	5	127	93.7%
Telco	3 and 1	24 and 18	100% (too few churners to rely on)
Bank	2	219	2.7%
Bank	0	782	14.8%
Bank	3	485	68.2%
Bank	1	144	97.9%



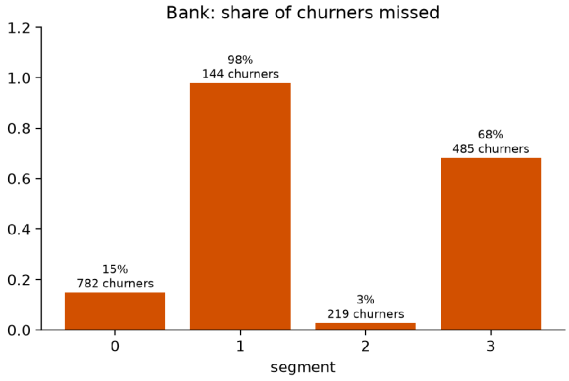


Figure 7. Share of churners missed, by segment (out-of-fold, cost-optimal threshold).

In Telco, missed churners are long-tenure customers (mean tenure 32.6 months against 11.9 for caught churners) with slightly lower monthly charges (68.2 against 77.9). Contract type matters most: 93.8% of one-year-contract churners and 100% of two-year churners are missed, against 22.9% of month-to-month churners. In the bank, missed churners are younger (mean age 38.6 against 48.3), more often active members (0.51 against 0.28) and hold fewer products (1.35 against 1.54). Misses are higher in France (44.3%) and Spain (45.2%) than in Germany (24.6%), and slightly higher for men (40.9%) than for women (32.9%).

In both datasets the missed churners have a mean predicted probability far below the threshold (0.19 in Telco, 0.15 in the bank, against 0.64 and 0.70 for caught churners). They look like loyal customers, so lowering the threshold would not recover them cheaply and the available features cannot identify them.

A hypothesis was tested for Telco: that long-tenure one-year-contract churners leave at contract renewal. Their tenures were not concentrated near multiples of 12 months (20.8% within one month of a multiple of 12, against about 25% expected by chance; n = 130). The test is inconclusive, because the data holds the current contract but not its start date, so tenure is not time since contract start. There is no evidence of a renewal effect, and the data cannot answer the question. A company would need contract start and end dates, complaint and usage data, and time-to-event models.

6. Survival analysis extension (Telco)

Telco churn is naturally time-to-event data: tenure is the duration and churn the event. Survival models change the question from "will this customer churn?" to "when?", and they let the cost matrix use a customer-specific value (RQ6). Only the Telco data is used because the bank data has no usable tenure-to-churn timeline.

6.1 Set-up

The time variable is tenure in months; the 8 training customers with tenure 0 were given 0.5 months. Non-churners are right-censored. Tenure and TotalCharges were removed from the

predictors, leaving 17. The training data holds 1,495 events among 5,634 customers; half of the churners left within 10 months, and the longest observed time is 72 months, the dataset's cap.

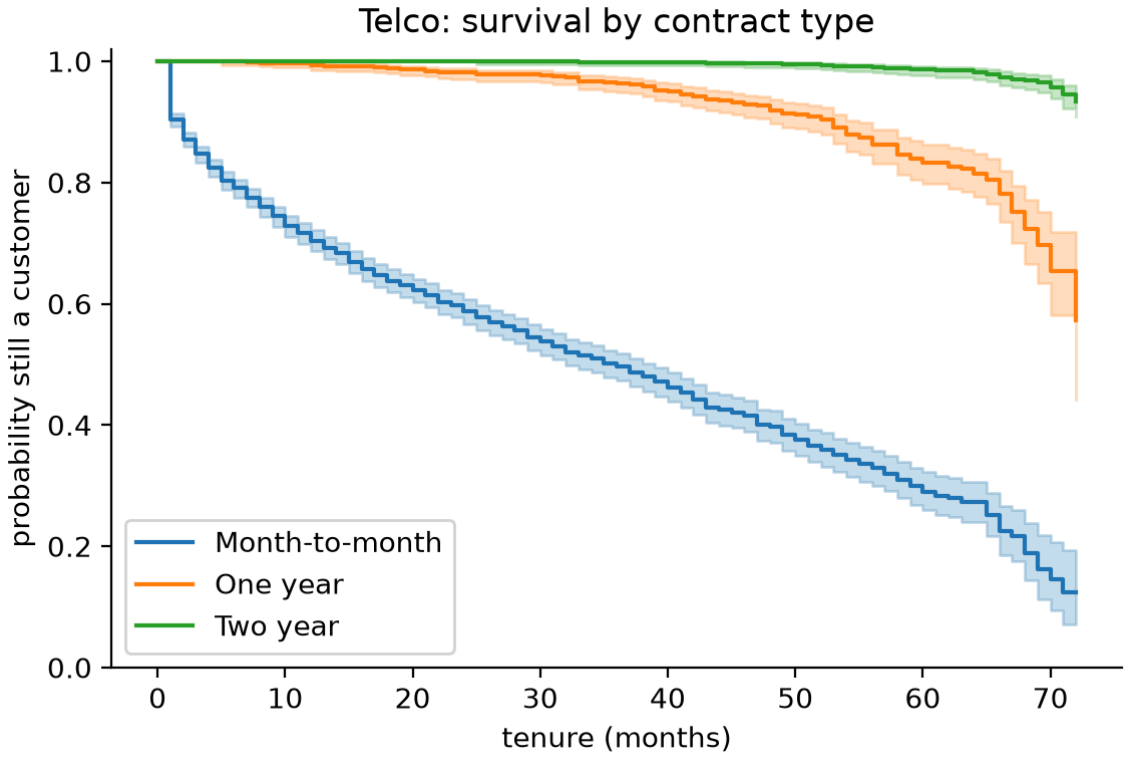


Figure 8. Probability of still being a customer by tenure, for each contract type (Kaplan-Meier curves with confidence bands, Telco training data).

6.2 Model comparison

Results are 5-fold cross-validation on the training data. A higher concordance index is better and a lower integrated Brier score is better.

Model	Concordance index (mean, sd)	Integrated Brier score (months 3-60)
No features (Kaplan-Meier)	n/a	0.175
Cox proportional hazards	0.867 (0.003)	0.084
XGBoost-AFT	0.865 (0.004)	0.082

Model	Concordance index (mean, sd)	Integrated Brier score (months 3-60)
Random Survival Forest	0.852 (0.004)	0.092
DeepSurv (neural Cox)	0.868 (0.004)	0.083

Cox, XGBoost-AFT and DeepSurv are tied within fold-to-fold variation; RSF is slightly lower. DeepSurv used one fixed, untuned configuration (two hidden layers of 32 units, dropout 0.1, weight decay 0.001, 200 epochs), like the other models. Its integrated Brier score (0.083) is level with Cox (0.084) and XGBoost-AFT (0.082). All models roughly halve the error of the no-features baseline, so the predicted survival probabilities carry real information. As in the classification comparison, the simple model matches the complex ones: a neural network adds nothing over the linear Cox model here.

6.3 Customer-specific lifetime value

The fixed value V = 777 assumed that every customer pays twelve months at the average bill. A customer-specific value replaces it:

V_i = monthly charge x expected number of active months in the next 12

The expected months come from the out-of-fold Cox survival curve, conditional on the customer's tenure so far, and the 12-month horizon keeps V_i comparable with 777. The mean V_i is 688.8 (range 41 to 1,425) and the mean expected activity is 10.7 of 12 months. Churners have a higher mean value than stayers (741 against 670) because they pay higher bills, so missing a churner is costlier than missing a typical customer.

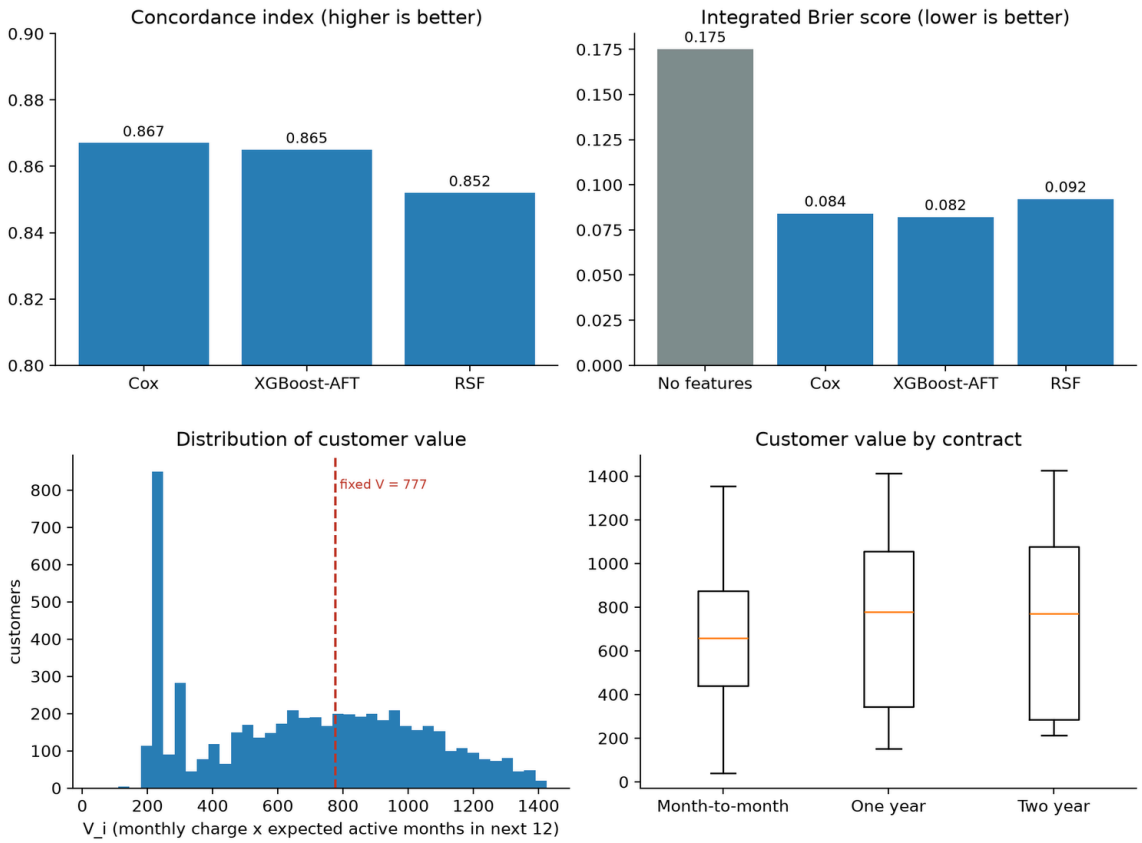


Figure 9. Survival model comparison (concordance index and integrated Brier score, top) and distribution of the customer-specific value V_i, overall and by contract (bottom).

6.4 Effect on the cost decision

The offer (78) and success rate (30%) are unchanged. With a fixed offer, each customer has a personal break-even probability, offer / (s x V_i): high-value customers are worth contacting at lower risk than low-value ones. Costs are per customer, on out-of-fold XGBoost probabilities.

	Fixed V = 777	Customer-specific V_i
Contact nobody	206.18	196.74
Single threshold 0.36	188.11	180.06
Best single threshold	188.11 (0.36)	179.60 (0.41)
Personal break-even rule	188.54	178.32

	Fixed V = 777	Customer-specific V_i
Customers contacted by the rule	1,897	1,673
Saving of the rule against contacting nobody	8.6%	9.4%

With fixed V the analysis reproduces the earlier results. With V_i the personal rule contacts fewer customers (1,673 against 1,897) and still costs less: it skips low-value customers for whom the offer is not worth its cost and contacts high-value customers at lower risk. The rule has no tuned parameter, while the single threshold was tuned on the same data. The gain over the best single threshold is small, about 1.3 per customer (0.7 percentage points), and should be presented as modest.

6.5 Limitations

The proportional-hazards assumption of the Cox model is only approximately met. A rank-based Schoenfeld-residual test rejects it at the 5% level for 9 of 21 features, and for two of them (Contract Two year and MultipleLines) after a Bonferroni correction for 21 tests. The comparison is not sensitive to this, because RSF and XGBoost-AFT, which do not assume proportional hazards, score within 0.015 concordance index of the Cox model, but V_i, which uses the Cox curves, should be read as an approximation. The AFT survival curve assumes a normal distribution of log lifetime with a fixed scale. V_i is an estimate, and it is revenue, not profit. Tenure is capped at 72 months. The test set was not used for the survival models.

7. Streamlit demonstration

A Streamlit application shows the whole chain for one customer. The user picks one of eight example customers, ordered from low to high risk, or edits any value, in either dataset. The page then shows:

1. **Churn probability** against the cost-optimal threshold, and whether the customer is flagged.
2. **SHAP explanation:** the eight features that push the risk up or down most.
3. **Segment:** the customer's SHAP-based segment, with its size, average churn and the suggested action.

4. **Recommended action:** an offer if the churn probability exceeds the break-even probability. For Telco, the break-even uses the customer-specific value from the survival model; for the bank it uses the assumed V = 1,000, so break-even is fixed at 0.33.
5. **Counterfactual options:** on request, up to three DiCE options that would bring a flagged customer below the threshold.

Two design choices are worth knowing. DiCE's random search sometimes proposes changes that remove something good (becoming inactive, dropping the credit card, switching off an add-on service) because the model's risk falls; the demo hides these as unrealistic retention actions, generates eight options and shows the first three realistic ones; the search uses a fixed seed (42), so the same customer always gets the same options. And segment assignment for a new customer uses SHAP values from the model fitted on all training data, whereas the segments themselves were built from out-of-fold SHAP values.

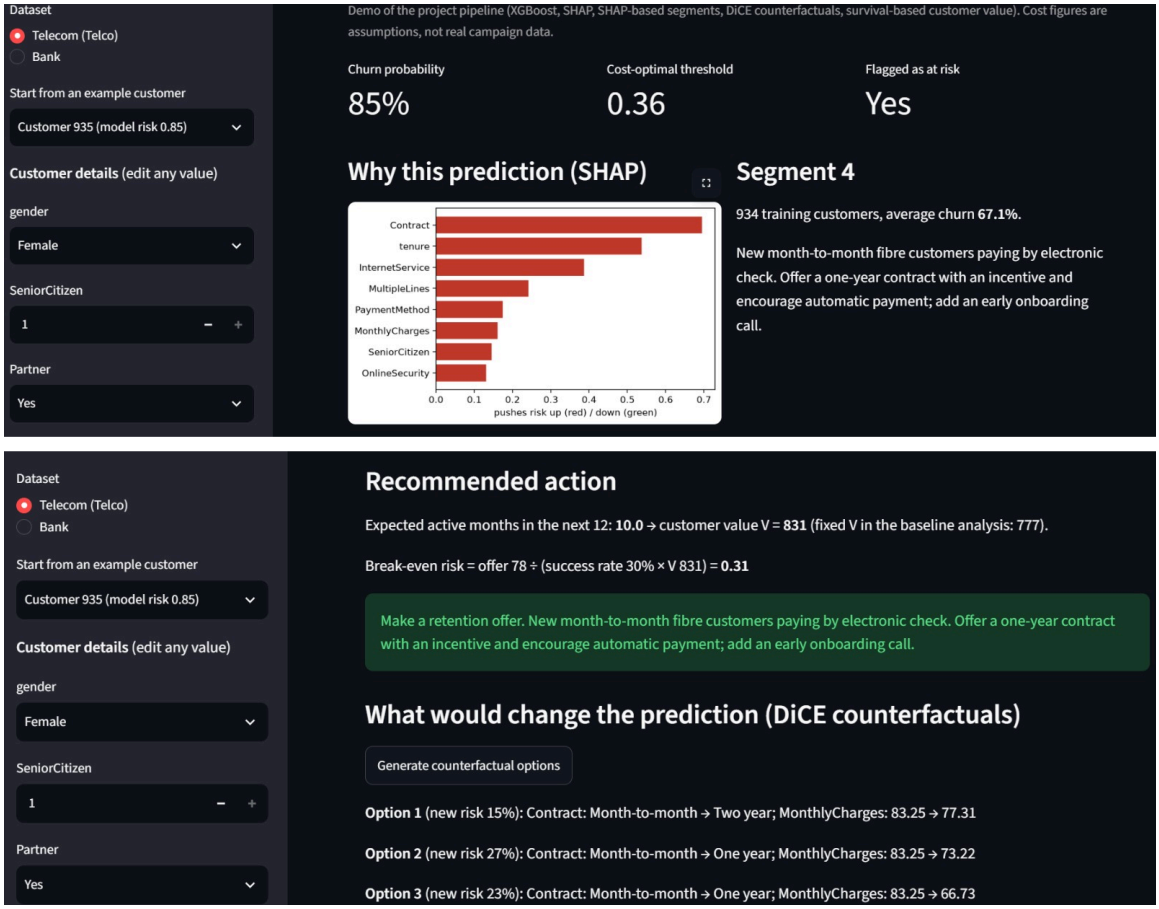


Figure 10. Streamlit demo, Telco: a high-risk customer with the recommended action and counterfactual options.

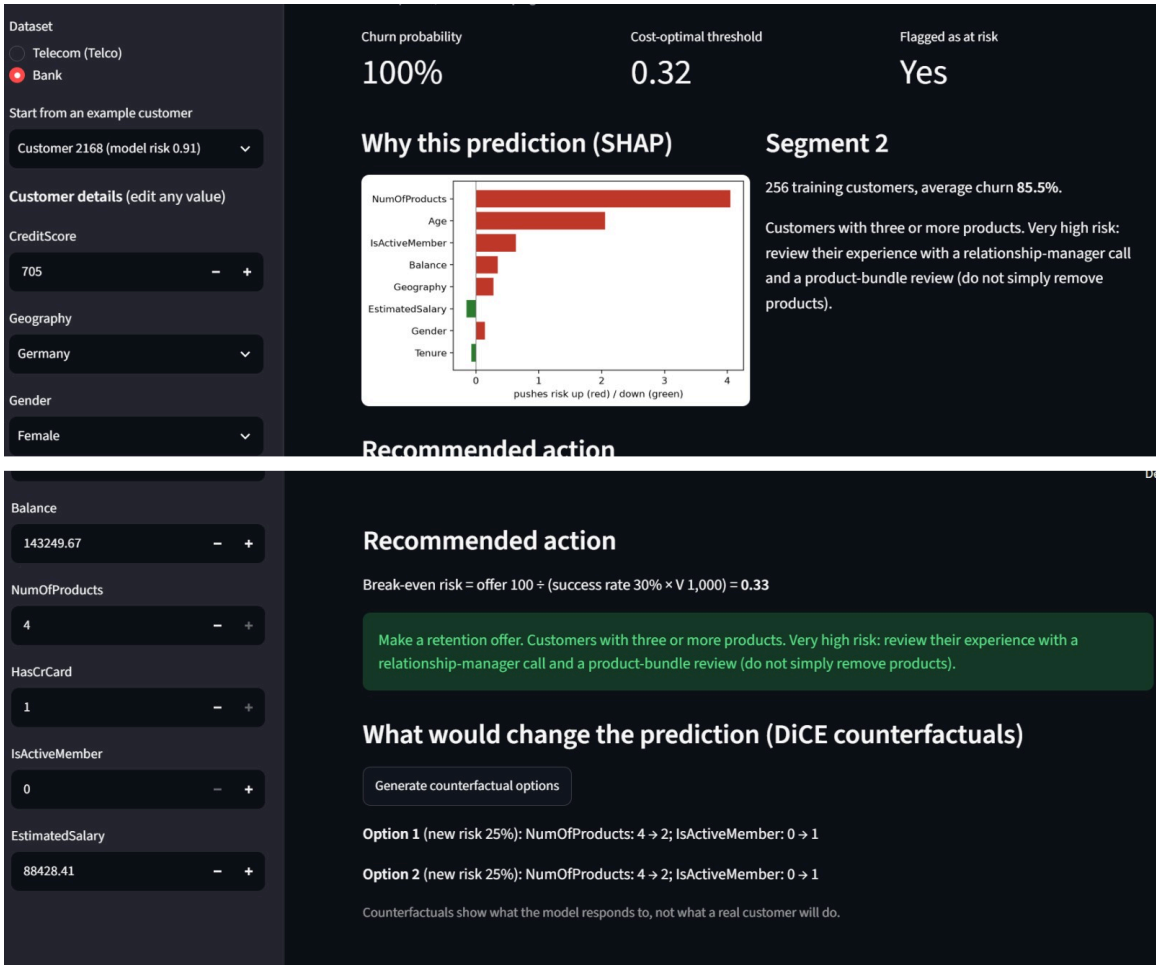


Figure 11. Streamlit demo, bank: a high-risk customer (four products) with the recommended action and counterfactual options.

One command builds the saved models (`python app/build_app_artifacts.py`, which also checks that the segments are reproduced exactly) and a second starts the application (`streamlit run app/streamlit_app.py`). Screenshots of low-risk and high-risk Telco customers, the two-year-contract effect and high-risk bank customers are in the project's `figures` folder.

8. Discussion and limitations

8.1 What the results say

Three findings hold in both datasets. First, model choice matters little once a reasonable boosting model is used: the top models are within noise on Telco, and EBM, a glass-box model, matches XGBoost on both datasets. Second, threshold choice matters more than model choice: choosing the threshold by cost saves about 9% against contacting nobody, while the choice among the top models changes PR-AUC by less than 0.01. Third, the segments that carry most of the churn are identifiable and actionable: in Telco, new month-to-month customers on fibre; in the bank, customers with three or more products and older single-product customers.

The explanations agree with each other. Contract type is the top SHAP driver in Telco, contract commitment is the stable counterfactual lever, and month-to-month customers show the highest churn in the exploratory analysis. The same agreement holds for number of products and age in the bank.

The error analysis qualifies these results. The model is best exactly where risk is most visible: it catches almost all churners in the riskiest Telco segment (98.9%) and in the riskiest bank segment (97.3%). What it misses are customers who look loyal, such as long-tenure Telco customers on one- or two-year contracts or younger, active bank customers. Catching them needs different data, not a lower threshold.

8.2 Limitations

- **Assumed costs.** No real campaign data exists. The offer cost, the success rate and, for the bank, the customer value are assumptions, and the sensitivity analysis shows that the saving ranges from 0% to about 35% depending on them. Both datasets use the same cost ratios, so their similar savings are partly by construction.
- **Associations, not causes.** SHAP and counterfactuals describe what the model has learned. A counterfactual shows what the model responds to, not what a real customer would do, and "becoming active" is an outcome that needs its own actions.
- **Risk-aware segments.** SHAP-based segments are built from a model trained on churn labels, so their better separation of churn against raw-feature clusters is partly by

construction. The Telco silhouette is low (0.23 to 0.26), so the segments are a practical partition of a continuum.

- **Small groups.** Bank segment 2 has 256 customers; the bank groups with three or four products are small (266 and 60); only four customers were analysed with counterfactuals.
- **Counterfactual options depend on the seed.** DiCE's random search is now seeded, so runs repeat exactly, but a different seed gives different add-ons and discounts; only the contract lever was stable across seeds and earlier runs in Telco.
- **Fairness.** Gender and geography appear as churn drivers in the bank data, and misses differ by group (France and Spain 44% to 45% against Germany 25%; men 41% against women 33%). A group check on out-of-fold predictions at the cost-optimal threshold shows that flag rates follow group churn rates, and that false-alarm rates are higher in higher-churn groups: bank women 0.137 against 0.071 for men, Germany 0.215 against about 0.07 in France and Spain, and Telco senior citizens 0.345 against 0.155 for others (Telco gender is balanced). These customers would receive more unnecessary offers. As a mitigation test, models were retrained without the sensitive attributes (Telco gender and senior-citizen status; bank gender and geography). In Telco this costs almost nothing (PR-AUC 0.660 against 0.663, expected cost 188.3 against 188.1), but the senior-citizen gap remains (false-alarm rate 0.369 against 0.193), because other features act as proxies. In the bank it narrows the gaps (false-alarm rate 0.097 for Germany against 0.086 for France and 0.104 for Spain; 0.095 for women against 0.091 for men) but weakens the model (PR-AUC 0.665 against 0.701, expected cost 187.5 against 185.7) and lowers recall for the highest-churn groups (Germany 0.578 against 0.754, women 0.571 against 0.671). Whether such attributes should be used to decide who receives offers is therefore a trade-off between fairness and accuracy, and a policy decision beyond this project.
- **Data limits.** The datasets are public and cross-sectional. Telco holds the current contract but not its start date, which is why the renewal hypothesis could not be tested. The bank data has no tenure-to-churn timeline, so survival analysis was applied to Telco only.
- **Survival analysis.** The proportional-hazards assumption is only approximately met (9 of 21 features reject it, two after Bonferroni correction), V_i is revenue rather than profit, and tenure is capped at 72 months.

- **Extensions run with fixed settings.** LIME and DeepSurv were run as additional checks with a single, untuned configuration each; neither changed a conclusion.

8.3 Future work

A pilot campaign should measure the real success rate of retention offers, which is the quantity the cost result depends on most. Further steps are to collect contract start and end dates and complaint and usage data, to compare other fairness methods (for example group-specific thresholds), and to extend survival analysis to the bank if timelines become available.

9. Conclusion

The project built one leakage-safe pipeline that predicts churn, explains it, segments customers by the reasons for their risk and recommends actions, and applied it to a telecom and a banking dataset. The research questions are answered as follows.

- **RQ1.** XGBoost, LightGBM, CatBoost and EBM perform within noise of each other (PR-AUC about 0.66-0.67 on Telco and 0.70-0.71 on the bank); logistic regression is competitive on Telco but weak on the bank (0.469). XGBoost was kept as the lead model.
- **RQ2.** Yes. Choosing the threshold by expected cost lowers cost by 8.7% (Telco) and 8.8% (bank) against contacting nobody on the untouched test set, and by about 1% against the default 0.5. The saving depends on how well offers work, from 0% to about 35% across the assumptions tested.
- **RQ3.** SHAP-based segments separate churn more widely than raw-feature segments (spread 0.654 against 0.498 in Telco, 0.808 against 0.191 in the bank), though this is partly because SHAP is risk-aware by construction. Their churn ranges from 1.7% to 67.1% (Telco) and 4.7% to 85.5% (bank), which supports targeting retention offers at a few segments.
- **RQ4.** In Telco the stable lever is contract commitment; in the bank the levers are activity and the number of products. These describe what the model responds to, not proven effects.
- **RQ5.** The model misses churners who look loyal: long-tenure customers on longer contracts in Telco (mean tenure 32.6 months) and younger, active customers in the bank

(mean age 38.6). Their predicted risk is far below the threshold, so they need new data, not a lower threshold.

- **RQ6.** A customer-specific value from survival analysis (concordance index 0.867) gives a modest improvement: a personal break-even rule saves 9.4% against 8.7% for the best single threshold, while contacting fewer customers.

Four further checks qualify these answers. The Cox proportional-hazards assumption holds only approximately, but a neural survival model (DeepSurv, concordance index 0.868) is no better than Cox (0.867), so the survival result does not depend on it. LIME agrees broadly with SHAP (3.3 to 3.7 of the top five features in common). False-alarm rates are higher for higher-churn groups (bank women, customers in Germany and Telco senior citizens), so these customers would receive more unnecessary offers; removing the sensitive attributes narrows these gaps in the bank at some cost in accuracy (PR-AUC 0.665 against 0.701) but not in Telco, so the project flags this as a policy matter and does not resolve it.

The main limitation is that the cost figures are assumptions. The most valuable next step is a pilot campaign that measures the real success rate of retention offers.

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Appendix A. AI-assistance disclosure

This project was developed with the assistance of an AI model (Claude, by Anthropic).

Appendix B. Reproducibility and repository

The code, notebooks, results and decisions log are in the project repository (github.com/sakethasakhamuri/churn-project). Python 3.12 is used with the packages in requirements.txt; seeds are fixed to 42.

Step	Notebook or file
Data cleaning	notebooks/01_data_cleaning
Exploratory analysis	notebooks/02_eda
Modelling, cost analysis, tuning, final test	notebooks/03_modelling
SHAP	notebooks/04_explainability_shap
Segmentation	notebooks/05_segmentation
Counterfactuals	notebooks/06_counterfactuals
Error analysis	notebooks/07_error_analysis
Survival analysis and lifetime value	notebooks/08_survival
Report figures	notebooks/09_report_figures
Proportional-hazards test, group check, LIME, DeepSurv	notebooks/10_extensions
Demo	app/build_app_artifacts.py, app/streamlit_app.py
Every methodological decision	notes/decisions_log.md (23 entries)

The notebooks were re-run from a clean start. All headline numbers reproduced exactly. The DiCE counterfactual search is seeded, so its options repeat between runs.